

Comprehensive SHAP-Based Explainability Analysis for Machine Learning-Based Bearing Fault Detection

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Abstract

This paper presents a comprehensive SHAP (SHapley Additive exPlanations)-based explainability analysis for bearing fault detection. We systematically compare two classic machine learning models, the Support Vector Classifier (SVC) and Logistic Regression with optimized hyperparameters, on a real-world bearing fault dataset called Case Western Reserve University (CWRU) dataset that has 10 fault classes. The optimized SVC model obtains 96.4% accuracy, with a ROC-AUC of 0.731 and a PR-AUC of 0.623. While the Logistic Regression model obtains 94.3% accuracy, with a ROC-AUC of 0.656 and a PR-AUC of 0.606. A thorough SHAP analysis conducted on both machine learning models with a 200-sample subset shows four important things: (1) SVM puts more weight on distribution shape features than Logistic Regression does on amplitude focus. SVC with nonlinear decision boundaries achieve superior ROC-AUC, while Logistic Regression provides interpretable linear relationships, (2) Feature Importance such as Inner Race faults depend mostly on mean amplitude, Outer Race faults depend on kurtosis, especially at high severity, and Ball faults depend on combined indicators, (3) Inner Race faults show decreasing mean importance with severity, while Outer Race faults show increasing kurtosis importance, and (4) The errors in the SHAP subset and the errors in the full test set occur at severity boundaries without mixing up fault locations. This suggests that feature engineering aimed at boundary discrimination could improve performance. The work encourage industry to use Logistic Regression for explainable baseline monitoring with mean as the main fault indicator and SVC for the most accurate detection with crest factor as the main fault indicator. The only features that appear in the top five for both models are mean and kurtosis. This shows that these features are important for all model architectures.

1 Introduction

Rolling-element bearing degradation still becomes a primary trigger of rotating-equipment stoppages in manufacturing [1]. Studies report that these faults contribute to roughly 40–51% of all rotating-equipment interruptions, and the resulting unplanned downtime often leads to substantial production losses, unexpected maintenance expenses, and heightened safety concerns [2]. The impact extends well beyond the failed component itself because prolonged downtime disrupts production flow, forces costly rescheduling, and places additional pressure on maintenance teams. In facilities operating high speed machinery or equipment with stringent safety requirements, a bearing malfunction can escalate into a serious hazard that puts workers at risk and threatens the integrity of surrounding assets and manufacturing infrastructure.

Traditional maintenance approaches have proven inadequate for modern industrial demands [3]. Reactive approaches that wait for components to fail expose facilities to unplanned

stoppages, production losses, and higher emergency repair expenses. Periodic preventive procedures that intended to reduce risk often result in components being replaced earlier than necessary, increasing maintenance expenditure while still leaving room for unexpected failures in parts that deteriorate faster than expected. Predictive maintenance creates a paradigm shift to enable maintenance decisions based on actual bearing condition assessment rather than arbitrary time intervals or reactive failure response [18].

Recent studies in the benchmark CWRU dataset achieved significant classification accuracy due to utilizing recent deep learning approaches. For example, CNN-based architectures reaching 99.91% [9], CNN-LSTM hybrid structures achieving 99.6% [11], and ResNet-50 transfer learning achieving 99.99% [6]. These great results represent the strong technical advancement for automated bearing fault detection. However, the use of AI-driven bearing fault detection in the real-world manufacturing plants remains limited. The main obstacle is not the accuracy itself but the absence of clear explanations behind

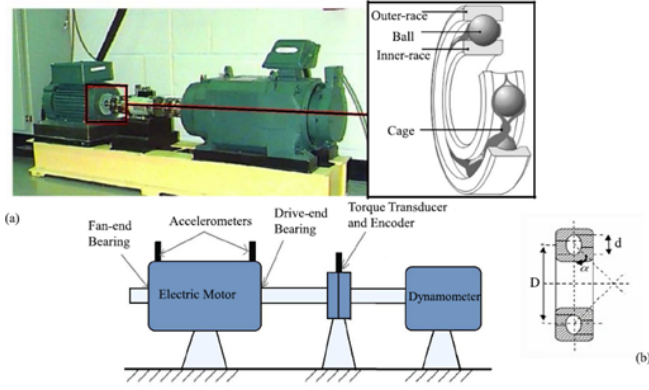


Fig. 1: CWRU Bearing Dataset Setup [13]

model decisions [14]. Maintenance personnel are unable to determine why a model flags a bearing as faulty or which signal features influence the output. This lack of clarity creates hesitation within maintenance teams, who are reluctant to rely on a black-box model that offers recommendations without revealing the reasoning behind its conclusions [14].

A fully transparent 94-96% accurate model that maintenance personnel understand and trust provides substantially greater industrial value than a 99% black-box model offering no insight into diagnosis mechanisms. This paper directly addresses this gap by providing explainability over marginal accuracy improvements. We employ comprehensive SHAP analysis on classical machine learning models selected for inherent interpretability while achieving competitive performances. The key contribution is systematic comparison revealing how fundamentally different model architectures (nonlinear SVC versus linear Logistic Regression) detect bearing faults through different feature prioritization strategies.

2 Related Work

2.1 Deep Learning Approaches

Recent deep learning approaches on detecting bearing faults focused on improving accuracy using the CWRU dataset [5]. Guo et al. put forward the Hierarchical Adaptive Deep CNN (ADCNN), which reached an accuracy of 97.7% by adapting convolutional layers to fault characteristics in a hierarchical way [9]. Kaya et al. employed Hybrid CNN-LSTM that achieves 99.6% by using two signals through separate spatial and temporal feature extraction branches before conducting fusion features [11]. This hybrid method utilizes the strength of CNN model in capturing local features while LSTM model captures temporal dependencies across the entire signal segment. Fine-tuning ResNet-50 on ImageNet for transfer learning got 99.99% accuracy by using features learned from huge image datasets [6]. The transfer learning pipeline converts 1D vibration signals to 2D spectrograms enabling ImageNet model application.

Even with these accuracy gains, interpretability continues to be difficult to obtain. Deep neural networks function in a way

that hides their internal reasoning, leaving the basis of their decisions unclear [14]. Real-world application requires to identify which parts of the signal drive the predictions or check the output model against established engineering insight. Only using accuracy is insufficient, and the absence of transparent reasoning reduces confidence in deploying such systems within manufacturing settings.

2.2 Explainable AI

SHAP (SHapley Additive exPlanations) is based on Shapley values from cooperative game theory providing model-agnostic explanations by quantifying contribution from data features to model predictions [12]. SHAP has a number of important benefits: (1) a strong theoretical basis in Shapley values that makes sure explanations are fair and strong; (2) the ability to work with any model type (linear, tree-based, neural networks) without needing to change the model; and (3) both global explanations (feature importance across all predictions) and local explanations (why specific predictions happened) through a unified framework [13]. SHAP values offer a straightforward way to interpret model behavior. A positive value explains that the feature makes the prediction closer to a certain class, while a negative value shows that the feature moves the prediction away from that class.

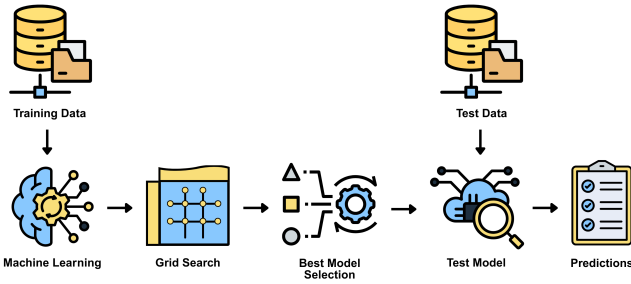
Recent detection of bearing faults shows that features identified by SHAP can achieve more than 90% classification accuracy with much smaller sets of features, sometimes only 5–10 features instead of the original 100+ features [17]. SHAP dependence analysis shows how features and targets are related, which helps practitioners figure out if features have linear or nonlinear relationships with fault probability. However, previous work focused on a single model or limited features without systematic comparison to model architectures and detailed investigation of how feature importance varies across fault types and severity levels [14].

3 Methodology

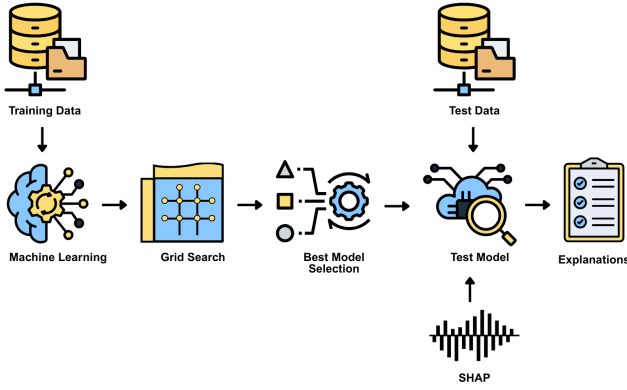
3.1 CWRU Bearing Dataset

The bearing dataset used in this paper is from The Case Western Reserve University (CWRU). The dataset contains 2,300 vibration samples and balanced across 10 fault classes with exactly 230 sample for each class [13]. The dataset composition includes on normal or healthy bearing condition and nine faulty bearing condition that represents three distinct fault positions which are inner race, outer race, and rolling element, at three different severity levels which are 0.007", 0.014", 0.021" fault diameters.

Fig. 1 shows experimental setup for CWRU bearing dataset utilizing a 2 HP Reliance Electric motor operating at four different load conditions (0 HP, 1 HP, 2 HP, 3 HP) corresponding to shaft speeds of approximately 1797, 1772, 1750, and 1730 RPM, respectively. The vibration data was acquired using accelerometers positioned at both motor drive end and fan end positions. In this paper, the data is sampled at 48 kHz with each signal segment containing 2,048 points representing 0.04



(a) Machine learning process without explainability



(b) Machine learning process with explainability

Fig. 2: The illustrations of machine learning approaches

seconds of continuous vibration acquisition. This sampling technique provides sufficient frequency resolution to capture bearing fault characteristic frequencies up to approximately 20 kHz where it is above typical bearing fault frequencies occurring in 500 Hz to 10 kHz range.

3.2 Feature Extraction

Nine statistical features in the time-domain were systematically extracted from each signal segment without requiring Fourier or wavelet transformation: maximum (peak value in segment), minimum (lowest value), mean (average amplitude), standard deviation (SD, amplitude variability measure), root mean square (RMS, effective amplitude), skewness (distribution asymmetry), kurtosis (impulsiveness measure), crest factor (peak divided by RMS) and form factor (RMS divided by mean). These features capture complementary signal aspects including amplitude behavior, variability, distribution shape, and impulsiveness characteristics critical for bearing fault diagnosis. Using Time-domain feature extraction can avoid computational complexity of Fourier or wavelet analysis and enable the deployment on resource-constrained IoT devices and embedded systems.

3.3 Data Preprocessing

From the total 2,3000 data samples, the data is split into training and test data with ratio 67.4% and 32.6%. We used stratified

random splitting to maintain fault class proportions in both training and test datasets. Hence this approach ensures that performance evaluation occurs on unbiased test sets with balanced fault class composition, preventing bias where one class dominates test data. Furthermore, the standard scaler normalization is used to transform each feature to zero mean and unit variance using training set statistics: $x_{norm} = (x - \mu_{train}) / \sigma_{train}$ where μ_{train} and σ_{train} represent mean and standard deviation computed exclusively from training data, respectively. Normalization of test data is used training set statistics to strictly prevent data leakage where test set statistics influence normalization applied during model training. This step is a standard approach to ensure rigorous generalization in performance evaluation.

3.4 Model Training and Hyperparameter Optimization

Support Vector Classifier (SVC) with RBF (Radial Basis Function) kernel was selected as the primary model due to demonstrated effectiveness for bearing fault classification and relative interpretability compared to deep neural networks [5]. SVC finds optimal nonlinear decision boundaries by mapping data into higher-dimensional spaces where classes become linearly separable through kernel transformation.

Hyperparameter optimization employed GridSearchCV with stratified 10-fold cross-validation. Specifically, we systematically optimized the regularization parameter $C \in \{1, 10, 45, 47, 49, 50, 51, 55, 100, 300, 500\}$ controlling the trade-off between margin maximization and training error minimization, and kernel parameter $\gamma \in \{0.01, 0.05, 0.1, 0.5, 1, 5\}$ controlling RBF kernel width. The 10-fold cross-validation process partitioned training data into 10 folds, trained on 9 folds while evaluating on held-out fold, repeating 10 times and averaging results. This process identified parameter combination ($C = 55$, $\gamma = 0.05$) providing best cross-validation performance. Probability estimation was enabled through `probability=True` parameter for SHAP compatibility, allowing SVC to output class probability estimates necessary for explanation generation.

Logistic Regression was selected as a linear baseline model enabling direct comparison with SVC's nonlinear approach. For multi-class classification (10 fault classes), we employed multinomial Logistic Regression with L-BFGS solver (limited-memory Broyden-Fletcher-Goldfarb-Shanno optimization) and maximum 1000 iterations. No regularization was applied, allowing the model to learn unrestricted linear relationships in feature space.

3.5 Explainability Implementation

SHAP value calculations employed different explainers optimized for each model architecture. The illustration on how machine learning process without explainability and with explainability is depicted in Fig. 2. For SVC non-linear model, we used KernelExplainer providing model-agnostic explanation through approximating Shapley value contributions by repeatedly masking feature subsets and measuring prediction

changes. For computational efficiency, we calculated SHAP values on a subset (200 test samples) rather than all 750 test samples. This subset approach maintains proportional representation of all fault types while reducing computation time.

For Logistic Regression, we employed LinearExplainer providing faster SHAP calculations specific to linear models through analytical Shapley value computation rather than approximation. Both explainers computed SHAP values for each feature relative to each of the 10 fault classes, generating three-dimensional arrays representing how each feature contributes to each class prediction.

4 Results and Analysis

4.1 Overall Model Performance Evaluation

The optimized SVC model achieved 96.4% classification accuracy on the 750-sample test set, correctly classifying 723 samples with only 27 misclassifications (3.6% error rate). Performance metrics included ROC-AUC (area under receiver operating characteristic curve) of 0.731 and PR-AUC (area under precision-recall curve) of 0.623, indicating solid discrimination between fault classes. In contrast, Logistic Regression achieved 94.3% accuracy (707/750 correct, 43 misclassifications), with ROC-AUC of 0.656 and PR-AUC of 0.606. The 2.1 percentage point accuracy difference between SVC and LR represents the performance tradeoff between nonlinear boundary learning and linear model transparency. Detail model comparison could be found in Table. 1 and confusion matrices is depicted in Fig. 3.

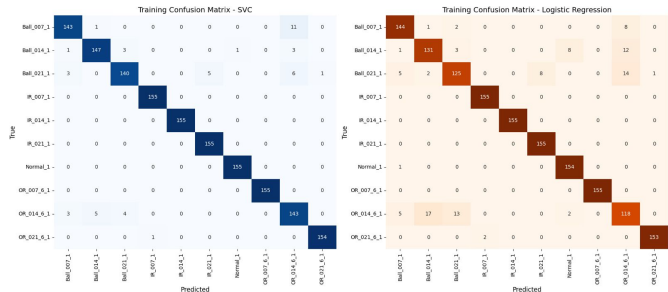


Fig. 3: Confusion Matrices

Per-class performance analysis revealed important variations across fault types and severities. SVC achieved perfect recall (100%) on several fault classes including IR0141, IR0211, OR00761, and OR02161 (no misclassifications on these classes), demonstrating complete discrimination for these faults. Performance was notably lower for Ball faults, with Ball0141 achieving 88% recall (66/75 correct samples) and Ball0211 achieving 89% recall (67/75 correct), indicating that ball faults exhibit more similar feature distributions to each other and to outer race faults, complicating discrimination at 0.014" and 0.021" severity levels.

Table 1 Model Performance Comparison on Test Data

Model	Accuracy	Errors	ROC-AUC	PR-AUC
SVC (Best)	96.4%	27	0.731	0.623
LR Baseline	94.3%	43	0.656	0.606

4.2 Global Feature Importance Analysis

SHAP analysis was performed using mean absolute SHAP values computed across the two hundred sample subset, providing an overall measure of how strongly each feature influenced the model’s predictions across all instances and fault types. Fig. 4 shows SHAP feature importance for SVC model. The top-5 most important features ranked by mean absolute SHAP value were: (1) crest factor with SHAP value 0.0508, the maximum influence feature capturing peak-to-RMS relationships; (2) minimum value with 0.0366 capturing extreme low signal values; (3) mean with 0.0291 measuring average amplitude; (4) skewness with 0.0256 measuring distribution asymmetry; (5) kurtosis with 0.0222 measuring distribution tail weight.

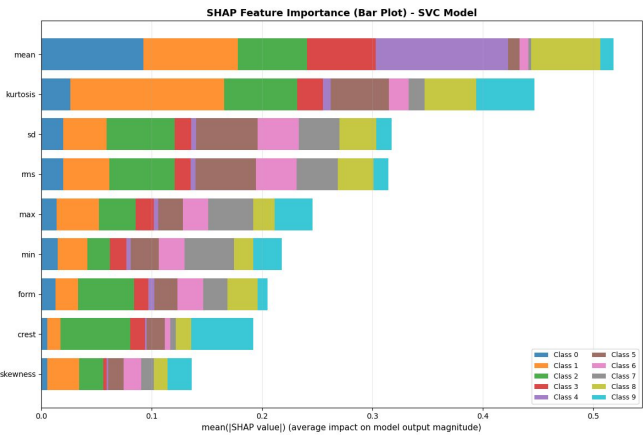


Fig. 4: SHAP Feature Importance Bar - SVC Model

Fig. 5 shows SHAP feature importance for Logistic Regression. The ranking differed dramatically: (1) mean with 1.739 dominating with massive influence magnitude; (2) standard deviation with 1.462 measuring amplitude variability; (3) RMS with 1.456 providing effective amplitude; (4) form factor with 1.200 measuring RMS over mean ratio; (5) kurtosis with 1.142. The magnitude difference in SHAP values between models (SVC maximum 0.0508 versus LR maximum 1.739, approximately 35-fold difference) reflects fundamental differences in how models utilize features. SVC with nonlinear decision boundaries achieves discrimination through complex feature interactions, so individual features contribute more modestly to predictions. Logistic Regression, constrained to linear boundaries, relies more heavily on individual feature magnitudes and directions, producing larger SHAP values.

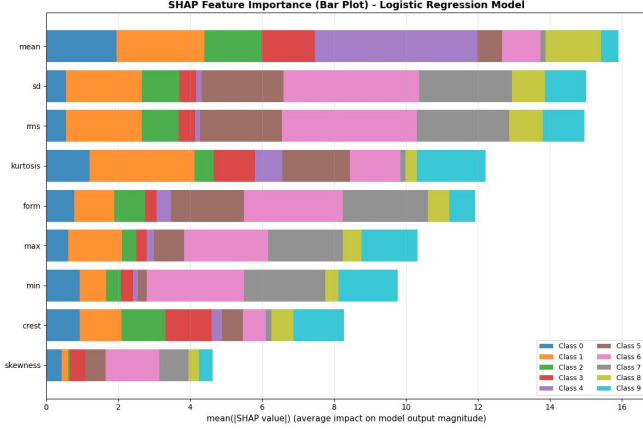


Fig. 5: SHAP Feature Importance Bar - Logistic Regression

Table 2 Top-5 SHAP-Ranked Features by Two Models

Rank	SVC Model	SHAP	LR Model	SHAP
1	Crest Factor	0.0508	Mean	1.739
2	Min Value	0.0366	SD	1.462
3	Mean	0.0291	RMS	1.456
4	Skewness	0.0256	Form Factor	1.200
5	Kurtosis	0.0222	Kurtosis	1.142

The main insight lies in the relative ordering of feature importance rather than the absolute SHAP values. When the Pearson correlation was computed between the SVC and Logistic Regression feature-importance rankings, the result was a correlation coefficient of -0.404, indicating significant negative correlation. This negative correlation means features ranked highly by SVC are ranked lower by Logistic Regression, and vice versa. Only mean and kurtosis appear in top-5 features for both models as in Table 2, suggesting these two features represent universally important fault indicators regardless of model architecture.

4.3 Fault Category Feature Importance Analysis

SHAP based feature importance evaluated separately for each fault category shows clear patterns that reflect the physical behavior of bearing degradation. For inner race (IR) faults, the mean amplitude emerged as the most influential feature across all fault severities. Specifically, IR0141 (0.014 in IR fault) demonstrated mean SHAP value of 0.1299, which is the highest single feature importance observed across all faults and features in the entire dataset. This dominance indicates that inner race faults are best characterized by mean amplitude changes in vibration signals. As inner race fault severity increases from 0.007 in to 0.021 in, mean importance decreases from 0.0828 to 0.0300, suggesting that larger inner race faults produce more consistent, predictable amplitude modulation patterns relative to baseline variability.

Outer race faults displayed a distinctly different pattern, with kurtosis emerging as the most influential feature. OR02161

(0.021 in OR fault) showed kurtosis as the dominant feature with SHAP value 0.0805, the highest kurtosis importance observed. As outer race fault severity increases from 0.007 in to 0.021 in, kurtosis importance increases dramatically from 0.0310 to 0.0805, indicating that larger outer race defects produce increasingly impulsive vibration signatures with sharper impacts and higher distribution tail weights.

Ball faults demonstrated mean amplitude as the dominant feature across all severity levels with score of 0.0812, 0.0494, 0.0440 for Ball0071, Ball00141, Ball0211, respectively, with relatively stable importance independent of severity. This pattern indicates ball defects produce consistent mean amplitude changes regardless of fault size, contrasting with both IR and OR severity-dependent patterns. Detail of most important feature on each fault category and their SHAP values is reported in Table 3.

Table 3 Most Important Feature on Each Fault Category and SHAP Values. Bold is the highest.

Fault Class	Top Feature	SHAP Value
IR0071	Mean	0.0828
IR0141	Mean	0.1299
IR0211	SD	0.0378
OR00761	SD	0.0381
OR01461	Mean	0.0501
OR02161	Kurtosis	0.0805
Ball0071	Mean	0.0812
Ball0141	Mean	0.0494
Ball0211	Mean	0.0440

4.4 Misclassification Analysis

Experiments using SHAP utilized 200-sample subset achieved 98% accuracy (196/200 correct) with only 4 misclassifications, higher than evaluation on test data with 96.4% accuracy. This difference reflects random sampling where 200 samples from 750 may select easy samples with clear fault signatures. Full test data evaluation with 96.4% accuracy represents the authoritative performance metric. Analysis of full test data misclassifications revealed important patterns, which is all misclassifications involved confusion between faults at similar severity levels. No systematic confusion occurred between faults at dramatically different severity levels (0.007" versus 0.021") or between different fault locations when fault severity was high (>0.014"). This pattern indicates the model achieves high discrimination between fault locations but struggles with severity-level discrimination, particularly at intermediate severity levels where feature distributions overlap significantly.

All misclassified cases occurred only at the boundary between severity levels (0.014" versus 0.021"). None of the errors involved mixing fault locations such as Ball, IR, or OR. Inspection of the feature values for these samples shows that they fall in intermediate regions where the distributions of the two severity levels strongly overlap. For instance as depicted in Fig.6, Ball0211 misclassified as OR01461 showed mean and kurtosis values intermediate between typical Ball0211 and

OR01461 ranges. The SHAP explanations for these misclassified instances revealed competing influences, where strong positive (red color) contributions supporting one class were counterbalanced by strong negative (blue color) contributions pointing to another class, ultimately leading the model to an incorrect decision.

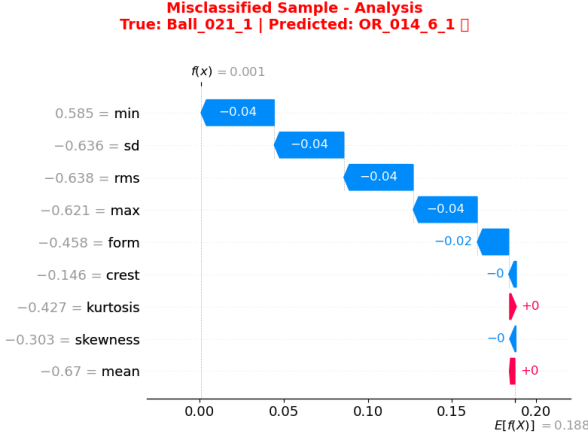


Fig. 6: Misclassified Samples

5 Discussion

5.1 Different Feature Importance on Two ML Models

The striking negative -0.404 correlation between SVC and LR feature importance reflects fundamental differences in how these models construct decision boundaries and utilize feature information. The Support Vector Classifier with an RBF kernel maps the input features into a higher dimensional space, allowing complex nonlinear class boundaries to be formed efficiently. In this space, the decision surface is determined by support vectors, which are the training samples positioned near the class margins. The nonlinear transformation and margin-based optimization enable SVC to exploit complex relationships including feature interactions and extreme value behaviors. Crest factor, which reflects the relationship between peak amplitude and RMS level, and the minimum value, which reflects extreme low points in the signal, serve as strong inputs for forming nonlinear decision boundaries. These indicators help the SVC model establish decision regions that respond effectively to distinctive extreme-value patterns, allowing clearer separation of different fault types.

Logistic Regression, which relies on linear decision boundaries, depends on features that offer clear separation through direct linear trends. Mean amplitude with a SHAP value of 1.739 aligns closely with bearing deterioration because damaged bearings generally show increased overall vibration levels. Standard deviation and RMS also support linear separability since both rise in a consistent manner as degradation progresses. Measures related to distribution shape, such as skewness and kurtosis, contribute less in this setting because

their diagnostic value often emerges only when nonlinear feature interactions are available.

5.2 Physical Implication of Fault Category Feature Importance

The fault category feature patterns align closely with the physical processes that generate each kind of bearing defect. For inner race faults, the prominence of the mean amplitude arises from the mechanics of a rotating defect. As the bearing completes each cycle, the rolling elements pass over the damaged inner race surface, producing repeated pulses. These pulses create periodic changes in overall vibration energy, and the signal envelope rises and falls with each contact event. This modulation is captured most directly in the mean amplitude, which explains its dominant influence in identifying inner race faults. Severity-dependent decrease in mean importance ($0.1299 \rightarrow 0.0300$) indicates that larger defects produce more stable and predictable modulation patterns.

Outer race dominance by kurtosis corresponds to the behavior of a fixed defect. When rolling elements strike a damaged outer race surface, they generate impulsive bursts rather than changes in average amplitude. The defining feature is the sharp, high-energy impacts. Kurtosis, which reflects tail heaviness and sensitivity to extreme values, captures this impulsiveness directly. Larger defects produce stronger impacts and therefore higher kurtosis levels, explaining the severity-dependent increase in kurtosis importance ($0.0310 \rightarrow 0.0805$).

Ball faults involve combined mechanisms in which defects on the rolling element generate impacts as the element moves through the load zone. The damaged ball contacts both the inner and outer race surfaces in a repeating manner, producing multiple sets of impacts tied to its position and rotation. This mixed, multi-frequency activity influences both mean amplitude through increased overall energy and kurtosis through the impulsive nature of the impacts, giving the two features comparable importance.

5.3 Severity Boundary Misclassifications and Implications

The discovered severity-dependent feature evolution patterns offer insights for condition-based predictive maintenance. Inner race faults show decreasing mean importance with increasing severity (0.1299 for $0.014''$ and 0.0300 for $0.021''$). This pattern suggests that very small inner race defects ($0.007''$) create subtle, variable modulation that is difficult for linear models to detect, hence higher relative mean importance for slightly larger defects ($0.014''$). As defects grow larger ($0.021''$), modulation becomes more consistent and predictable, requiring less relative reliance on mean amplitude as a distinguishing feature.

Outer race faults show opposite behavior with increasing kurtosis importance at higher severity (0.0310 for $0.007''$ and 0.0805 for $0.021''$). This pattern reflects how outer race defects accumulate energy as they grow larger, producing increasingly sharp impact signatures. Small defects ($0.007''$) produce relatively subtle impacts (low kurtosis), while large defects

(0.021") produce very sharp impacts (high kurtosis). This monotonic increase in kurtosis with severity suggests that monitoring kurtosis trends could provide early prediction of outer race defect progression. A bearing showing stable low kurtosis might suddenly show increasing kurtosis as outer race degradation progresses.

Ball faults show relatively stable feature patterns across severity levels, suggesting ball defect progression may involve consistent characteristic features rather than evolving patterns. This stability might reflect that ball defects impact bearing behavior similarly regardless of size—once a defect exists, it creates impacts throughout bearing operation, with larger defects creating proportionally larger impacts rather than qualitatively different signatures.

These severity patterns enable development of maintenance strategies adapting to fault progression. For inner race faults, early detection through mean amplitude monitoring followed by stability assessment at later stages. For outer race faults, progressive monitoring of kurtosis trends to predict defect growth. For ball faults, detection through combined indicators with less emphasis on progression tracking. Such fault-specific strategies improve maintenance planning efficiency and safety.

5.4 Model Selection Guidelines for Industrial Deployment

The comparative analysis of SVC (96.4% accuracy) versus Logistic Regression (94.3% accuracy) with fundamental feature divergence provides clear guidance for model selection in industrial applications, which depends on application requirements and priorities.

For explainability-oriented baseline monitoring, Logistic Regression provides clear benefits. Its linear boundaries make every feature coefficient directly interpretable, allowing maintenance teams to understand and verify model outputs without advanced ML expertise. The 94.3% accuracy, though slightly below the SVC result, is often sufficient for early warning and escalation rather than final maintenance calls.

For maximum detection accuracy in safety-critical applications, SVC's 96.4% accuracy provides meaningful improvement through nonlinear pattern recognition. The 2.1 percentage point improvement might seem modest, but for bearings operating at high speed or safety-critical loads, preventing a single failure could justify deployment of more complex models. SVC remains relatively simple compared to deep learning, deployable on modest computing hardware while offering substantial accuracy advantage.

For improved accuracy and robustness, combining SVC and Logistic Regression in an ensemble similar to what has been done by Shi et al., [19] is a promising direction. The negative correlation in their feature rankings shows that each model captures different aspects of the signal. A simple voting scheme can merge SVC's strong pattern recognition with LR's more stable, interpretable checks, yielding performance near 96.4%. Cases where the two models disagree can be flagged for manual review, offering a practical form of uncertainty indication.

6 Limitation and Future Work

Our analysis was limited to the CWRU laboratory dataset. Cross-dataset validation on Real industrial datasets that we will collect would confirm whether discovered feature patterns generalize across different bearing manufacturers, operating conditions, and degradation mechanisms. Next, as only time-domain features were examined in this paper, a structured SHAP comparison using frequency-domain and time-frequency features is recommended. Including bearing characteristic frequency information may strengthen severity separation. Moreover, complementary techniques including attention mechanism visualization showing which signal portions models prioritize, Grad-CAM and saliency maps identifying critical signal regions, and uncertainty quantification indicating prediction confidence would enhance interpretability further. For the next work, SVC and LR's feature divergence (-0.404 correlation) suggests ensemble voting combining both models could leverage complementary strengths, potentially exceeding 96.4% accuracy while improving robustness and confidence estimates.

7 Conclusion

This SHAP-based explainability approach shows that clarity and strong predictive performance can be achieved together in bearing fault diagnosis. The SVC model reaches 96.4% accuracy by capturing complex nonlinear patterns, while Logistic Regression reaches 94.3% accuracy with fully transparent linear approach. The SHAP evaluation produced four key findings. First, the SVC and LR feature correlation of -0.404 shows that the two models rely on contrasting decision processes. Second, each fault position displays distinct features, with mean amplitude highlighting inner race faults, kurtosis highlighting outer race faults, and both indicators contributing to ball fault detection. Third, feature importance changes with severity, offering a basis for maintenance actions guided by feature trends. Fourth, errors were confined to the severity boundary region, pointing to clear priorities for refining feature design.

Only mean and kurtosis emerged as consistently influential across both models, indicating that these should serve as core monitoring indicators in practical deployments. As diagnostic tools play a growing role in shaping maintenance actions and supporting safety-critical decisions, clarity in model behavior becomes as important as numerical performance for responsible industrial use. The physical aligned feature patterns identified in this paper create a solid basis for explainable, edge-ready bearing diagnostic systems suitable for Industry 4.0 predictive maintenance environments.

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